

Artificial Intelligence

Lesson 2: Anatomy of AI Systems

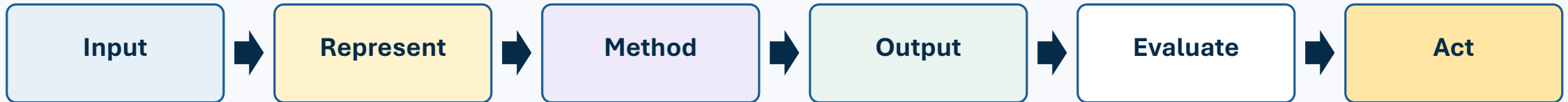
From input

→ computation

→ decision

→ evaluation

**Look under the hood
before trusting the
output.**



Retrieval warm-up

Lesson 1 ended with trust. Lesson 2 starts with system anatomy.

An AI system gives a convincing answer.



Before trusting it, what would you want to know about what happened behind the interface?

What information went in?

How was it represented?

What method produced the answer?

What checks happened before action?

Think-pair-share: name one invisible part of the system that could change the output.

Today's mission

By the end of Lesson 2, students should be able to...

1

Identify the major components of an AI system pipeline

2

Explain how inputs are transformed into outputs

3

Relate search, learning, and inference to system components

Main idea: the model is only one part of the system.

A model is not the whole system

A model can generate an output; a system decides what the output means.



AI model

The computational component that maps some representation to a prediction, route, label, score, or text.

AI system

- Collects and prepares inputs
- Runs one or more AI methods
- Applies rules, thresholds, and guardrails
- Supports a human or automated action

Reliable AI depends on the entire pipeline, not just the intelligence of the model.

Running scenario: drone route planning

How does a messy mission become a recommended route?

**Mission: get a drone from launch point
to target location without violating
constraints.**

Terrain

Weather

No-fly zones

Battery

Goal

Risk limits

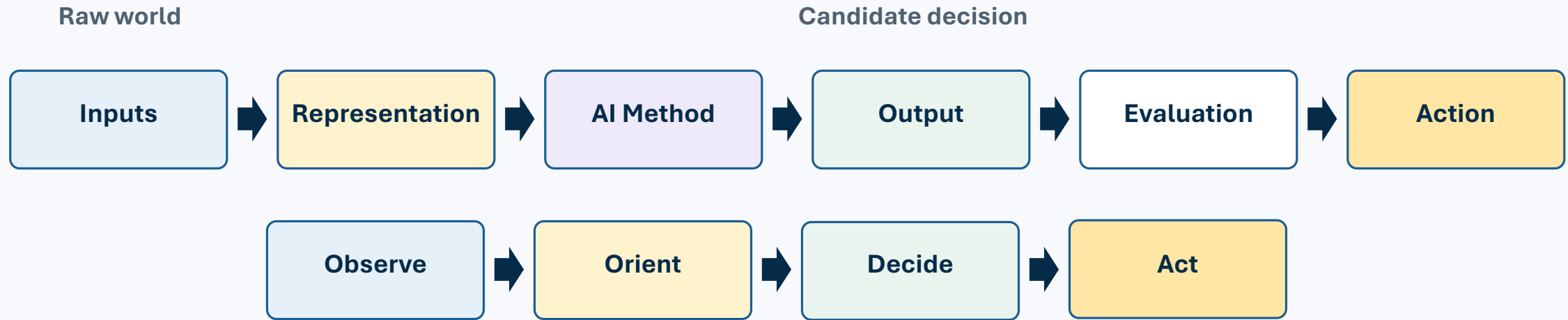


**Recommended route
+ explanation
+ confidence
+ constraints checked**

Question: what information would be dangerous to omit?

The AI system pipeline

A reusable map for understanding almost any AI system.

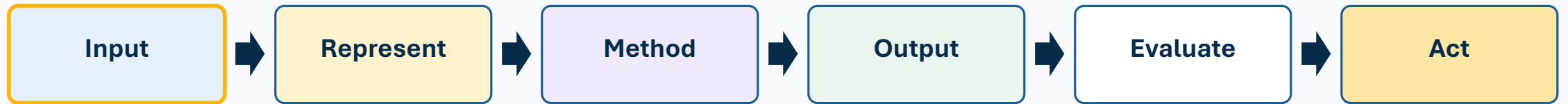


Feedback and monitoring: observe outcomes, detect failures, adjust future behavior

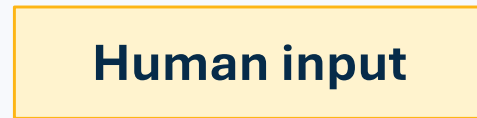
We will build this pipeline one stage at a time.

Stage 1: Inputs

AI systems begin with data, goals, constraints, and instructions.



location, images, radar,
telemetry



goals, prompts, tasking



maps, rules, prior
cases

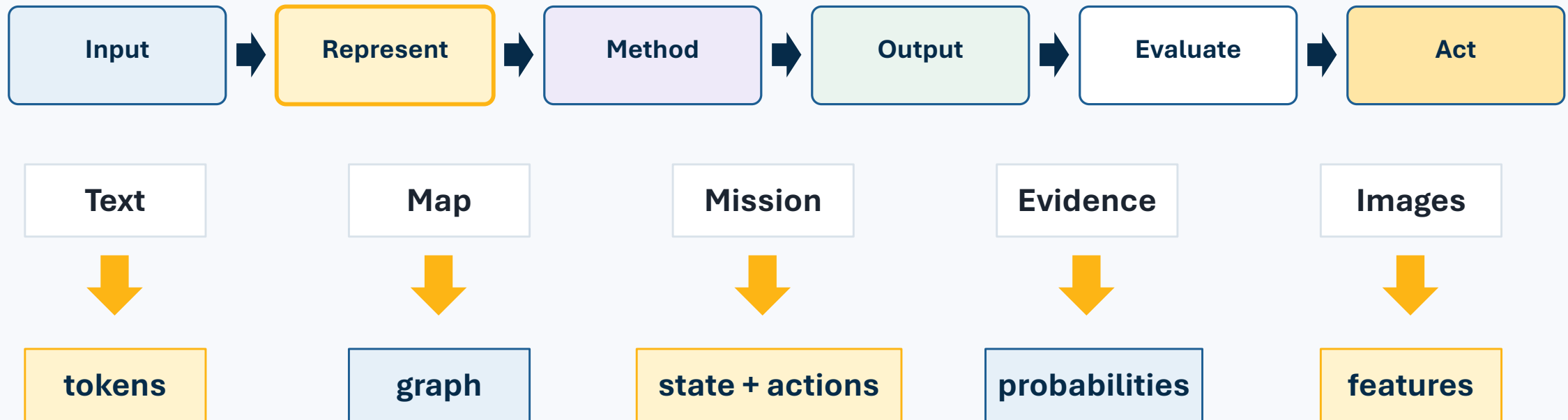


time, safety, policy,
resources

Available data is not always relevant, correct, current, or sufficient.

Stage 2: Representation and processing

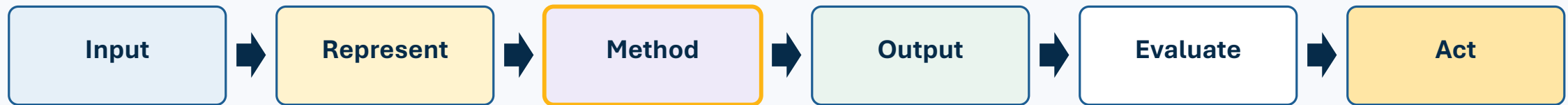
Raw information must become something the system can compute with.



Representation choices can make a problem easy, hard, or impossible to solve well.

Stage 3: The AI method

Different problems call for different engines.



Search

Find a path or sequence of actions.

route planning, scheduling

Learning

Find patterns from examples.

classification, prediction

Inference

Reason from evidence under uncertainty.

diagnosis, belief updating

This course teaches how to choose, apply, and evaluate these methods.

Same pipeline, different engine

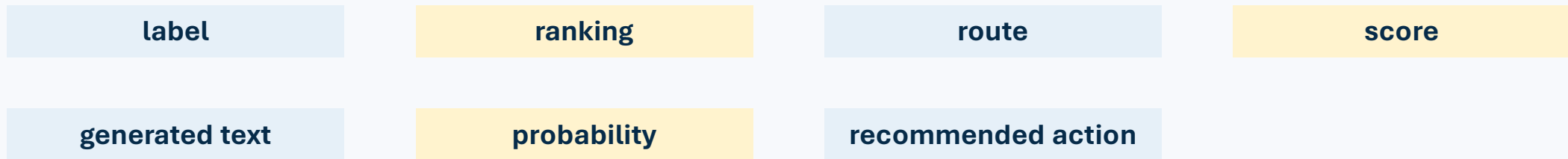
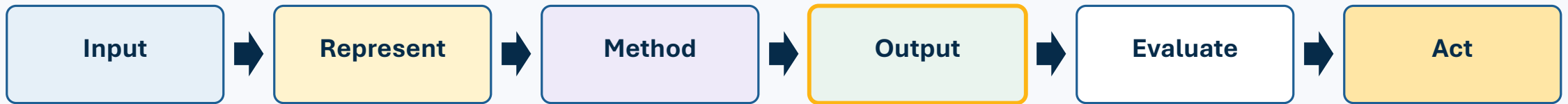
The anatomy stays similar even when the AI method changes.

System	Input	AI method	Output
Route planner	map + constraints	search	recommended path
Image classifier	pixels	learning	class label + score
Fault diagnosis	symptoms + evidence	inference	likely cause

Prompt: what changes across systems? What stays the same?

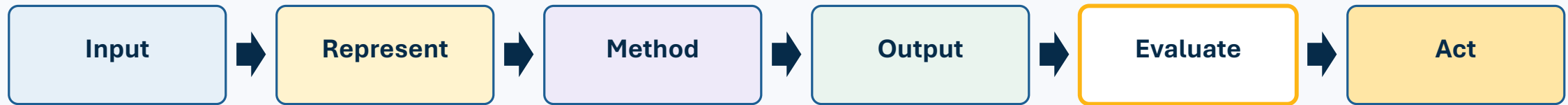
Stage 4: Outputs

An output is not automatically a decision.



Stage 5: Evaluation and guardrails

The system must decide whether the output is good enough to use.



Validation

Does it satisfy known facts?

Thresholds

Is confidence high enough?

Constraints

Does it violate rules or safety limits?

Human review

Should a person approve?

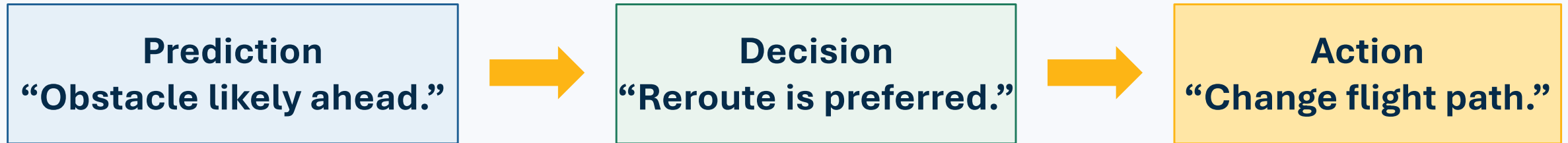
Fallback

What happens if it fails?

Lesson 1 rule: outputs must be evaluated, not blindly trusted.

From recommendation to action

The higher the stakes, the stronger the safeguards.

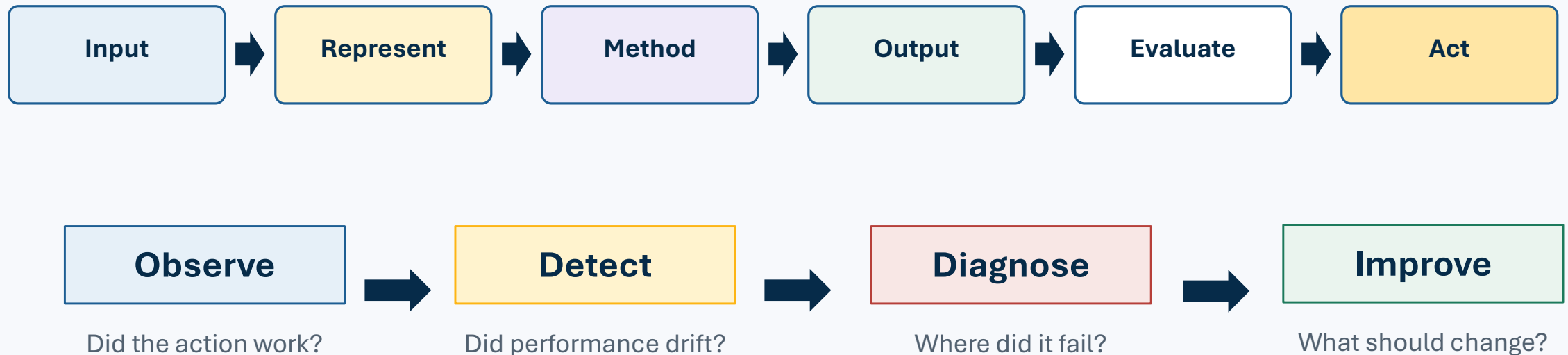


More autonomy  More consequence

Low-stakes outputs may only need light checking.
Operational actions require validation, constraints, and accountability.
A system should know when not to act.

Feedback and monitoring

AI systems operate in a changing world.



Feedback is how systems catch errors, update assumptions, and remain useful over time.

Where did the failure begin?

Diagnose the pipeline stage, then propose a safeguard.

Case A

A sensor reports the wrong position.

**Likely stage:
Input**

Case B

A map is represented without restricted airspace.

**Likely stage:
Representation**

Case C

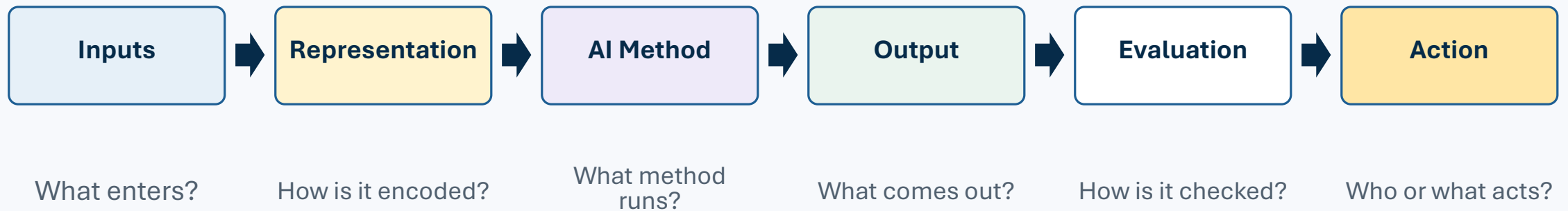
The system optimizes only speed and ignores safety.

**Likely stage:
AI method / objective**

Group task: for one case, name one safeguard that would reduce the risk.

Synthesis: anatomy of an AI system

Rebuild the pipeline from memory.



Reliable AI is a system design problem.

The quality of the final action depends on every stage before it.

Exit Ticket

Apply the pipeline to an AI system of your choice.

Choose one AI application and identify:

Input

AI method

Output

Evaluation step

Example answer: “Drone route planner: terrain + constraints → search → route → check safety and no-fly-zone compliance.”

Next class: Artificial Intelligence Workshop — change inputs, observe outputs, find failures.